

multi-Chemokine Receptor Analysis

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2023-04-17

A0A6B0S3N8

First Domain

Sequence [66-360]: VGQIEAQLLPPLYSLVFIFGFVGNLLVVLILINCKKLKSMTDIYLLNLAIS-DLLFLLTMPF WAHYAADQWVFGNVMCKFFTGLYHIGYFGGIFFIILLTIDRYLAIVHAVFALKARTVT-FGVVTSGVTWVAVFASLPGII FIKSLEEHSYGACAPYFPLGWKNFHTIMRSILGLVPLLMIVCYSGI-IKTLRLCRNEKKKKHKAURLIFVIMIVYFLFWA PYNIVLLLSTFQEFFGLSNCKSSSQLDQAMQVTETL-GLTHCCINPIIYAFVGEKFRRLSTFFRKHIAKHLCKQ

Homologies

Psi-blast search against Swiss-Prot:

First 5 Full Homologues

##	Receptor Name	Organism	Percent Identity	Query Cover
## 1	C-C chemokine receptor type 2	Macaca mulatta	84.07	100%
## 2	C-C chemokine receptor type 2	Rattus norvegicus	82.03	100%
## 3	C-C chemokine receptor type 2	Mus musculus	81.69	100%
## 4	C-C chemokine receptor type 5	Bos taurus	83.56	99%
## 5	C-C chemokine receptor type 2	Homo sapiens	84.48	93%

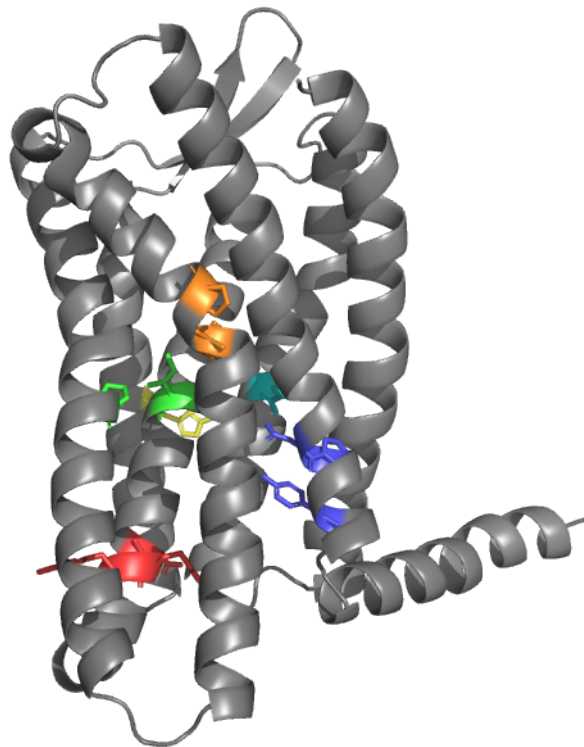
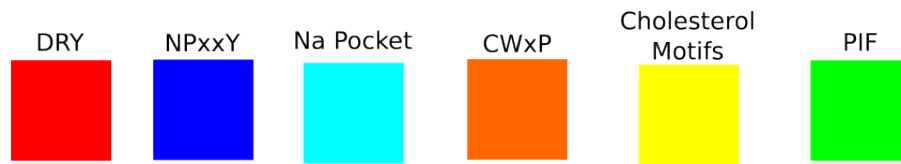
First 50 Unique Homologues

##	Receptor Name	Percent Identity
## 1	C-C chemokine receptor type 2	84.07
## 2	C-C chemokine receptor type 2	82.03
## 3	C-C chemokine receptor type 2	81.69
## 4	C-C chemokine receptor type 5	83.56
## 5	C-C chemokine receptor type 2	84.48
## 6	C-C chemokine receptor type 5	80.20
## 7	C-C chemokine receptor type 5	80.20
## 8	C-C chemokine receptor type 5	79.19
## 9	C-C chemokine receptor type 5	77.59
## 10	C-C chemokine receptor type 5	79.93
## 11	C-C chemokine receptor type 5	81.21
## 12	C-C chemokine receptor type 5	80.87
## 13	C-C chemokine receptor type 5	80.54
## 14	C-C chemokine receptor type 5	80.87
## 15	C-C chemokine receptor type 5	80.54
## 16	C-C chemokine receptor type 5	80.87
## 17	C-C chemokine receptor type 5	80.87
## 18	C-C chemokine receptor type 5	80.87
## 19	C-C chemokine receptor type 5	80.20
## 20	C-C chemokine receptor type 5	80.87
## 21	C-C chemokine receptor type 5	80.87
## 22	C-C chemokine receptor type 5	79.53
## 23	C-C chemokine receptor type 5	80.87
## 24	C-C chemokine receptor type 5	80.54
## 25	C-C chemokine receptor type 5	77.59
## 26	C-C chemokine receptor type 5	80.87
## 27	C-C chemokine receptor type 5	80.54
## 28	C-C chemokine receptor type 5	80.54
## 29	C-C chemokine receptor type 5	80.20
## 30	C-C chemokine receptor type 5	80.54
## 31	C-C chemokine receptor type 5	80.54
## 32	C-C chemokine receptor type 5	80.54
## 33	C-C chemokine receptor type 5	80.54
## 34	C-C chemokine receptor type 5	80.20
## 35	C-C chemokine receptor type 5	80.20
## 36	C-C chemokine receptor type 5	80.54
## 37	C-C chemokine receptor type 5	79.87
## 38	C-C chemokine receptor type 5	80.20
## 39	C-C chemokine receptor type 5	80.54
## 40	C-C chemokine receptor type 5	80.54
## 41	C-C chemokine receptor type 5	80.20
## 42	C-C chemokine receptor type 5	80.20
## 43	C-C chemokine receptor type 5	78.86
## 44	C-C chemokine receptor type 5	78.52
## 45	C-C chemokine receptor type 5	78.52
## 46	C-C chemokine receptor type 5	80.20
## 47	C-C chemokine receptor type 5	79.53
## 48	C-C chemokine receptor type 5	78.86
## 49	C-C chemokine receptor type 3	61.20
## 50	C-C chemokine receptor type 1	61.22

The similarity of the 50th match: 61% identity.

Structure

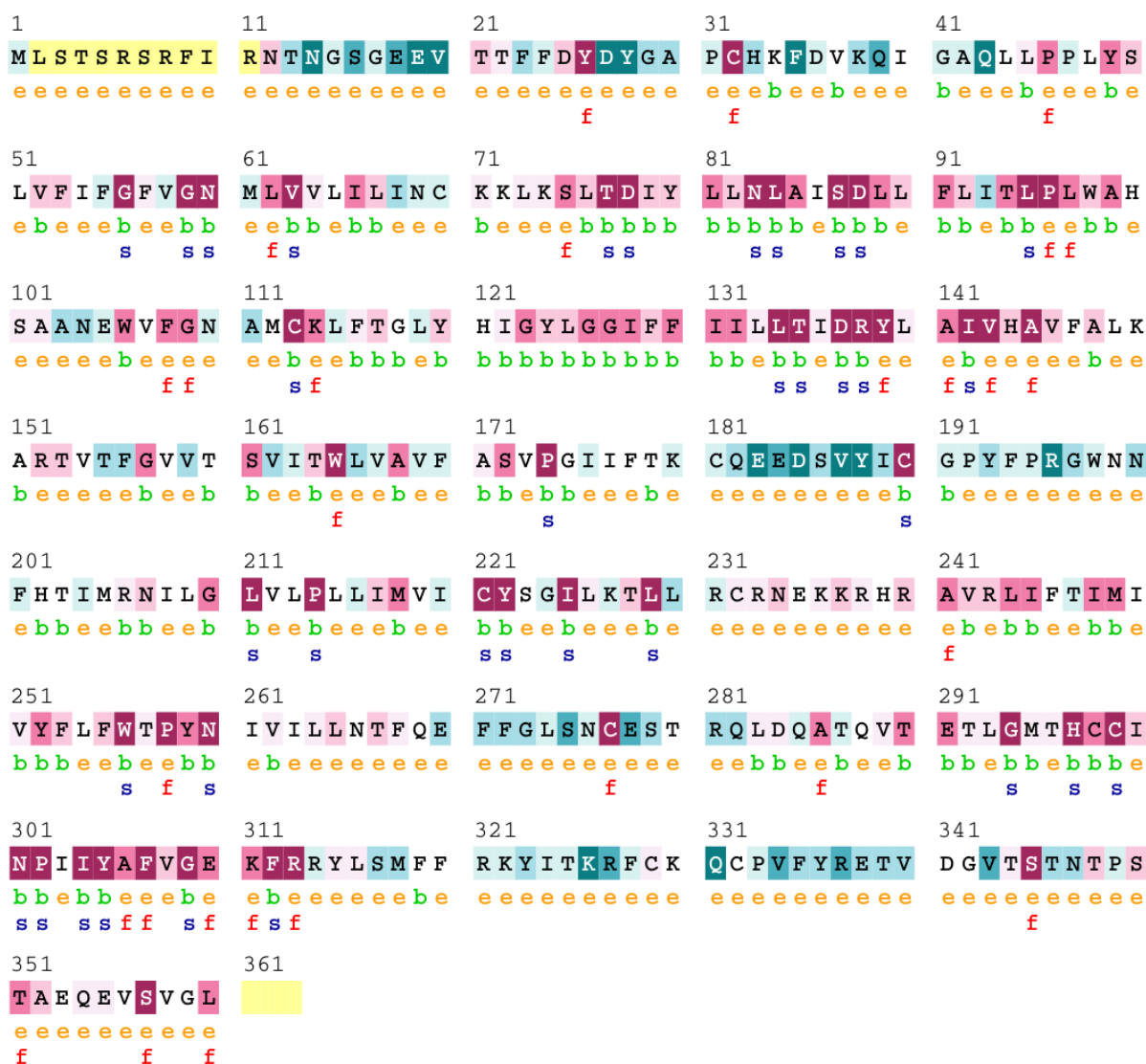
Motifs



Conservation

Conservation of the protein from SwissProt with highest homology to the first domain: C-C chemokine receptor type 2, UniProt ID: O18793

Conservation in the homomorph



Pairwise alignment with homolog Looking at the conserved sites predicted by ConSurf, and cross-referencing to our first domain sequence:

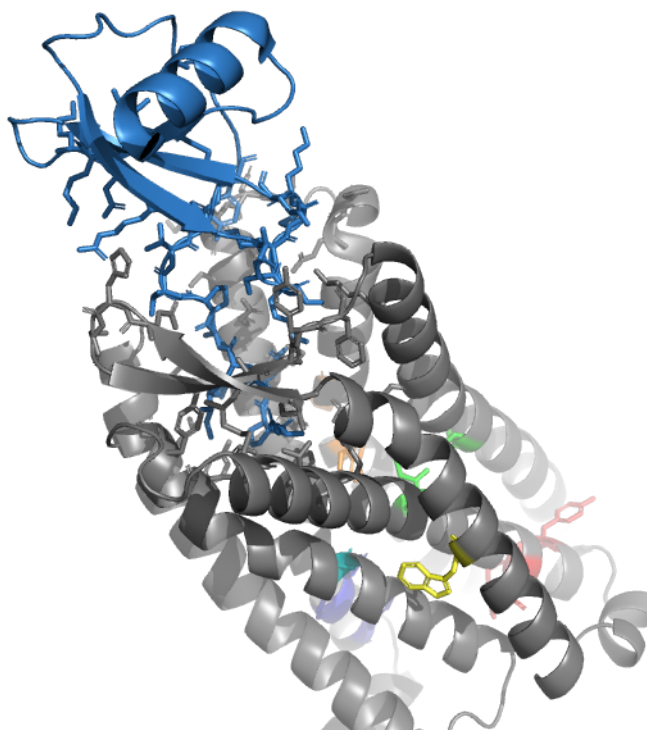
first_domain	1	-----VGQIEAQLLPPLYS	14
			
018793	1	MLSTSRSRFIRNTNGSGEEVTTFDYDYGAPCHKFDVKQIGAQLLPPLYS	50
first_domain	15	LVFIFGFVGNLLVVLILINCKKLKSMTDIYLLNLAISDLLFLLTMPFWAH	64
		. .	
018793	51	LVFIFGFVGNMLVVLILINCKKLKSLTDIYLLNLAISDLLFLITLPLWAH	100
first_domain	65	YAADQWVFGNVMCKFFTGLYHIGYFGGIFFIILLTIDRYLAIVHAVFALK	114
		. .	
018793	101	SAANEWVFGNAMCKLFTGLYHIGYLGGIFFIILLTIDRYLAIVHAVFALK	150
first_domain	115	ARTVTFGVVTSGVTWVAVFASLPGIIFIKSLEEHSYGACAPYFPLGWKN	164
		. .	
018793	151	ARTVTFGVVTSVITWLVAVFASVPGIIFTKCQEEDSVYICGPYFPRGWN	200
first_domain	165	FHTIMRSILGLVLPLLVMIVCYSGIIKTLLRCRNEKKKKHAVRLIFVIMI	214
		. .	
018793	201	FHTIMRNILGLVLPLLIMVICYSGILKTLLRCRNEKKRHRAVRLIFTIMI	250
first_domain	215	VYFLFWAPYNIVLLLSTFQEFFGLSNCKSSSQLDQAMQVTETLGLTHCCI	264
		. .	
018793	251	VYFLFWTPYNIVILLNTFQEFFGLSNCESTRQLDQATQVTETLGMTHCCI	300
first_domain	265	NPIIYAFVGEKFRRYLSTFFRKHIAKHLCKQ-----	295
			
018793	301	NPIIYAFVGEKFRRYLSMFFRKYITKRFCQCPVFYRETVDGVTSTNTPS	350
first_domain	296	-----	295
018793	351	TAEQEVSVGL	360

Binding ligands of homolog

7XA3 is with natural ligand (CCL2), 6GPX is a non-natural ligand (MK-0812),

Ligand superimposition of domain and homolog with ligand

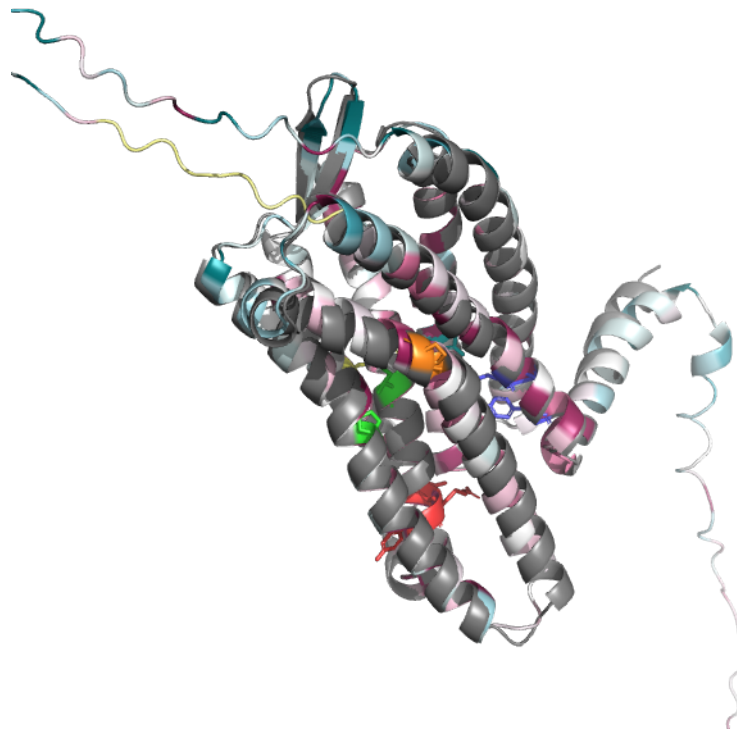
RMSD with the ligand-bound receptor: 1.391 Å



Of note, the extracellular side has some different amino acid residues, even though the pairwise alignment is almost identical. This might be important in deciding if the binding ligand is in fact CCL2.

Superimposition with best homolog+ConSurf

RMSD: 0.556



* Both are AlphaFold predicted structures

Second domain

Sequence[412-709]:LNVRQIAARLLPPLYSLVFIFGFVGNMLVVLILINCKKLKSMTDIYLLNLAISLLFIITIPF
WAHYAADQWVFGNTTCQLFTGFYFIGYFGGIFFIILLTIDRYLAIVHAVFALKARTVTFGAATSVVTWVVAV
FASLPGIIFTKSQKEGSRHTCSPHFPSSQYHFWKNFQTLKIVILGLVLPLLVMIVCYSGIIKTLLRCRNEKKKH
KAVRLIFVIMIVYFLFWAPYNIVLLLSTFQEFGGLNNCSGSNRLDQAMQVTETLGMTHCCINPIIYAFVGEKFR
NYLLRFFRKYFASRF

Homologies

Psi-blast search against Swiss-Prot:

First 5 Full Homologues

##	Receptor Name	Organism	Percent Identity
## 1	C-C chemokine receptor type 5	Bos taurus	99.66
## 2	C-C chemokine receptor type 5	Allochrocebus lhoesti	87.58
## 3	C-C chemokine receptor type 5	Allochrocebus solatus	87.58
## 4	C-C chemokine receptor type 5	Cercocebus atys	87.58
## 5	C-C chemokine receptor type 5	Pongo pygmaeus	87.25
##	Query Cover E value		
## 1	100% 0.0		
## 2	100% 0.0		
## 3	100% 0.0		
## 4	100% 0.0		
## 5	100% 0.0		

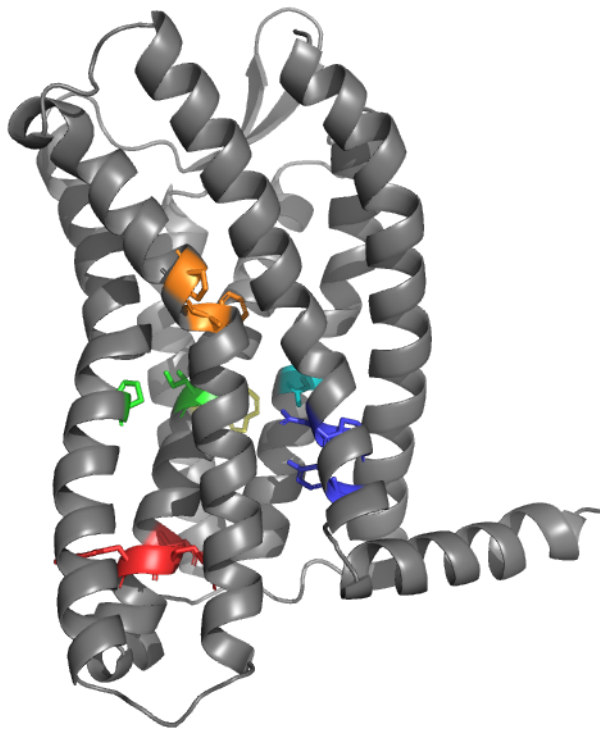
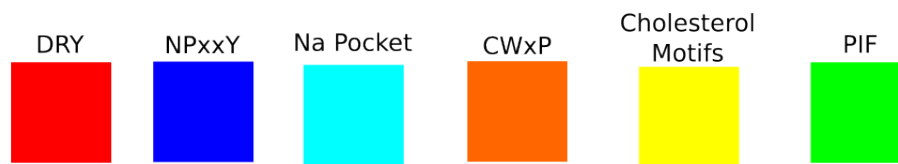
First 50 Unique Homologues

##	Receptor Name	Percent Identity
## 1	C-C chemokine receptor type 5	99.66
## 2	C-C chemokine receptor type 5	87.58
## 3	C-C chemokine receptor type 5	87.58
## 4	C-C chemokine receptor type 5	87.58
## 5	C-C chemokine receptor type 5	87.25
## 6	C-C chemokine receptor type 5	87.25
## 7	C-C chemokine receptor type 5	87.25
## 8	C-C chemokine receptor type 5	86.91
## 9	C-C chemokine receptor type 5	87.25
## 10	C-C chemokine receptor type 5	87.54
## 11	C-C chemokine receptor type 5	86.91
## 12	C-C chemokine receptor type 5	83.56
## 13	C-C chemokine receptor type 5	86.91
## 14	C-C chemokine receptor type 5	86.91
## 15	C-C chemokine receptor type 5	86.91
## 16	C-C chemokine receptor type 5	81.54
## 17	C-C chemokine receptor type 5	86.91
## 18	C-C chemokine receptor type 5	86.91
## 19	C-C chemokine receptor type 5	86.91
## 20	C-C chemokine receptor type 5	86.58
## 21	C-C chemokine receptor type 5	86.91
## 22	C-C chemokine receptor type 5	86.91
## 23	C-C chemokine receptor type 5	86.91
## 24	C-C chemokine receptor type 5	86.91
## 25	C-C chemokine receptor type 5	86.91
## 26	C-C chemokine receptor type 5	86.91
## 27	C-C chemokine receptor type 5	86.91
## 28	C-C chemokine receptor type 5	86.58
## 29	C-C chemokine receptor type 5	86.58
## 30	C-C chemokine receptor type 5	86.58
## 31	C-C chemokine receptor type 5	86.58
## 32	C-C chemokine receptor type 5	86.24
## 33	C-C chemokine receptor type 5	86.24
## 34	C-C chemokine receptor type 5	86.58
## 35	C-C chemokine receptor type 5	86.24
## 36	C-C chemokine receptor type 5	85.91
## 37	C-C chemokine receptor type 5	85.91
## 38	C-C chemokine receptor type 5	83.56
## 39	C-C chemokine receptor type 5	83.56
## 40	C-C chemokine receptor type 5	83.50
## 41	C-C chemokine receptor type 5	82.55
## 42	C-C chemokine receptor type 5	81.54
## 43	C-C chemokine receptor type 5	80.13
## 44	C-C chemokine receptor type 2	78.11
## 45	C-C chemokine receptor type 5	83.50
## 46	C-C chemokine receptor type 2	77.36
## 47	C-C chemokine receptor type 2	76.35
## 48	C-C chemokine receptor type 2	78.80
## 49	C-C chemokine receptor type 1	61.82
## 50	C-C chemokine receptor type 3	60.94

The similarity of the 50th match: 61% identity.

Structure

Motifs



Conservation

Conservation of the protein from SwissProt with highest homology to the second domain: C-C chemokine receptor type 5, UniProt ID: Q2HJ17

Conservation with homolog



Pairwise alignment with homolog

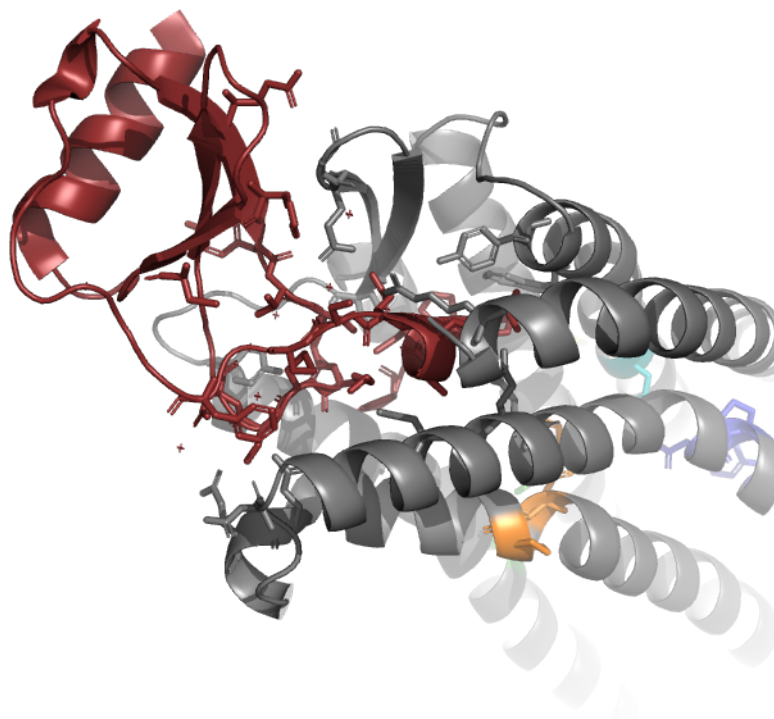
second_domain	1	-----LNVRQIAARLLPLYSLVFIFGFVGNML	28
Q2HJ17	1	MDYQTSTPLYDIDYGMSEPCQKLNVRQIAARLLPLYSLVFIFGFVGNML	50
second_domain	29	VVLILINCKKLKSMTDIYLLNLAISDLLFIITIPFWAHYAADQWVFGNTT	78
Q2HJ17	51	VVLILINCKKLKSMTDIYLLNLAISDLLFIITIPFWAHYAADQWVFGNTM	100
second_domain	79	CQLFTGFYFIGYFGGIFIIILLTIDRYLAIVHAVFALKARTVTFGAATSV	128
Q2HJ17	101	CQLFTGFYFIGYFGGIFIIILLTIDRYLAIVHAVFALKARTVTFGAATSV	150
second_domain	129	VTWVAVFASLPGIIFTKSQKEGSRHTCSPHPSSQYHFWKNFQTLKIVI	178
Q2HJ17	151	VTWVAVFASLPGIIFTKSQKEGSRHTCSPHPSSQYHFWKNFQTLKIVI	200
second_domain	179	LGLVLP LLVMIVCYSGIIKTLLRCRNEKKKKHKA VRLIFVIMIVYFLFWAP	228
Q2HJ17	201	LGLVLP LLVMIVCYSGIIKTLLRCRNEKKKKHKA VRLIFVIMIVYFLFWAP	250
second_domain	229	YNIVLLLSTFQEFG LNCSGSNRLDQAMQVTETLGMTHCCINPIIYAFV	278
Q2HJ17	251	YNIVLLLSTFQEFG LNCSGSNRLDQAMQVTETLGMTHCCINPIIYAFV	300
second_domain	279	GEKFRNYLLRFFRKYFASRF-----	298
Q2HJ17	301	GEKFRNYLLRFFRKYFASRFCKGCPVFQGEAPERVSSVYTRSTGEQEISV	350
second_domain	299	-- 298	
Q2HJ17	351	GL 352	

Binding ligands of homolog

5UIW is with natural ligand (CCR5), 6AKX, 6AKY is a non-natural ligand (Compound 21, 34),

Ligand superimposition of domain and homolog with ligand

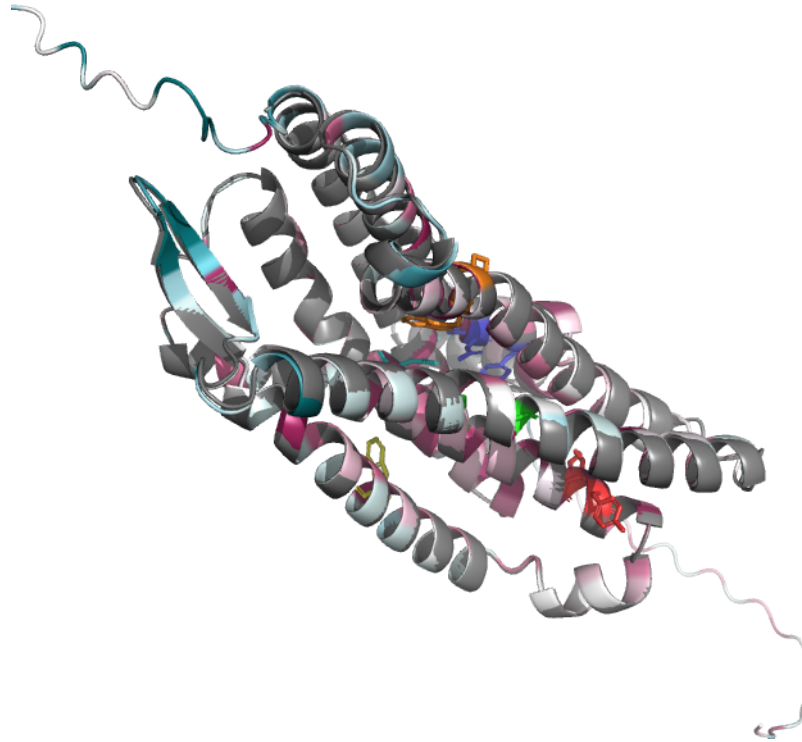
RMSD with the ligand-bound receptor: 0.506 Å



Of note, the extracellular side has some different amino acid residues, even though the pairwise alignment is almost identical. This might be important in deciding if the binding ligand is in fact CCL2.

Superimposition with ConSurf'd best homolog

RMSD: 0.317



* both structures are AlphaFold predictions

Pairwise alignment

Genomic sequence

15